Bikram Pandit

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Professional Summary

PhD candidate specializing in multi-robot control, reinforcement learning, and sim-to-real transfer, with 5 publications at top-tier peer-reviewed robotics conferences (ICRA, CoRL), including an Oral and a Spotlight, a 1st-place finish in the DARPA TIAMAT Challenge, and policies deployed on 6+ real robot platforms spanning bipeds, quadrupeds, wheeled-bipeds, and humanoids.

Education

Oregon State University — **Corvallis, Oregon**
PhD in Artificial Intelligence — *Sept 2021 – Present*
Research focus: large-scale robot learning, decentralized control, sim-to-real transfer, and loco-manipulation systems

Rungta College of Engineering and Technology — **India**
Bachelor's in Computer Science and Engineering — *Aug 2013 – June 2017*

Awards and Honors

- **1st Place – DARPA TIAMAT Challenge** — **2026**
  Developed robust Spot policies for high-speed exploration, fall recovery on unpredictable terrain, and obstacle avoidance in cluttered settings; integrated with an autonomous planner for sequential task execution.
- **Best Paper Award Finalist (Cognitive Robotics)** — **ICRA, 2024**
  Learning Vision-Based Bipedal Locomotion for Challenging Terrain.
- **2nd Place – AgAID Institute Hackathon** — **2023**
  Competed against 32 teams from 8+ universities in a Microsoft-sponsored hackathon.

Research Experience

Dynamic Robotics and AI Lab, Oregon State University — **Corvallis, Oregon**
Research Assistant — *Sept 2021 – Present*

- Developed decentralized multi-robot control policies for cooperative payload transport and object manipulation, scaling across bipeds, quadruped-arm teams, and humanoids (CoRL 2024, ICRA 2026).
- Contributed to the design of vision-based RL policies for bipedal locomotion over challenging terrain using onboard camera and proprioception, without explicit state estimation (ICRA 2024, Best Paper Finalist).
- Contributed to masked-training strategies improving policy robustness to sensor dropout across locomotion and manipulation tasks (CoRL 2024).
- Co-developed a hyperparameter-free energy-minimization method for RL-based robot control, cutting energy use without sacrificing task performance (CoRL 2025, Oral).
- Built simulation and training infrastructure on IsaacLab, MjLab, and MuJoCo; contributed to open-source codebases including IsaacLab, RSL-RL, MuJoCo, and Weights & Biases.
- Deployed learned policies on 6+ real robot platforms, including Cassie, Digit, Spot, D1H wheeled-biped, Unitree Go2, and H1, with accelerated onboard inference for real-time control.

RAI Institute (formerly Boston Dynamics AI Institute) — **Cambridge, Massachusetts**
Robotics Research Intern — *Sept 2024 – Jan 2025*

- Developed multi-instance Isaac Sim infrastructure supporting heterogeneous robot articulations for scalable data generation and policy training across humanoid loco-manipulation research.
- Built unified imitation learning pipelines consolidating multiple specialized controllers into a single deployable policy.
- Validated natural walking and running behaviors in simulation and on real humanoid hardware.

Publications

- **Learning Multi-Humanoid Pickup and Transport via Decentralized Object-Centric Control** — **Under review**
  Object-centric control abstraction unifying single- and multi-robot humanoid pickup, transport, and handoff, validated on Digit V3 humanoids.
- **Multi-Quadruped Cooperative Object Transport: Learning Decentralized Pinch-Lift-Move** — **ICRA, 2026**
  Gripperless pinch-lift-move coordination for quadruped-arm teams of 2–10, validated on Unitree Go2 quadrupeds with D1 arms.
  Project Page · Code
- **Non-conflicting Energy Minimization in RL Based Robot Control (Oral)** — **CoRL, 2025**
  Hyperparameter-free energy optimization for RL control with sim-to-real validation.
  Project Page · Code
- **Learning Decentralized Multi-Biped Control for Payload Transport (Spotlight)** — **CoRL, 2024**
  Decentralized policy scaling rigid cooperative carrying to arbitrarily sized teams of bipeds sharing one carrier, validated on Cassie robots.
  Project Page · Code
- **Learning Vision-Based Bipedal Locomotion for Challenging Terrain (Best Paper Finalist)** **ICRA, 2024**
  Vision-conditioned sim-to-real locomotion robust to challenging terrain without explicit state estimation.
  arXiv
- **Simple Masked Training Strategies Yield Control Policies That Are Robust to Sensor Failure** — **CoRL, 2024**
  Robust policy training improving resilience to sensor dropout in real-world deployments.
  Paper

Patents

- **Systems and Methods for Decentralized Multi-Robot Cooperative Transport** — **Under review**

Industry Experience

MiCROTEC Inc.
Machine Learning Intern — *June 2023 – Aug 2023*

- Accelerated large-scale vision model training by 80% through optimized data pipelines and system-level performance tuning.
- Developed robust segmentation models addressing class imbalance and deployment constraints for production systems.
- Integrated model training, evaluation, and deployment into automated ML DevOps pipelines using Docker, CI, and experiment tracking.

Digital Product Labs
Senior Software Developer — *Dec 2017 – June 2021*

- Designed and deployed production machine learning systems across mobile, web, and cloud platforms.
- Integrated machine learning components into consumer-facing products, improving performance and scalability in production.
- Partnered with product and engineering teams to scope and ship AI-driven features end to end.